

MAT 362 HW09

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Due 04/10

Problem. 4.2.3. Show that a diffeomorphism $\varphi: S \rightarrow \bar{S}$ is an isometry if and only if the arc length of any parametrized curve in S is equal to the arc length of the image curve by φ .

Solution. todo □

Problem. 4.2.4. Use the stereographic projection to show that the sphere is locally conformal to a plane.

Solution. todo □

Problem. 4.2.7. Let V and W be (n -dimensional) vector spaces with inner products denoted by $\langle \bullet, \bullet \rangle$ and let $F: V \rightarrow W$ be a linear map. Prove that the following are equivalent:

- (a) $\langle F(v_1), F(v_2) \rangle = \langle v_1, v_2 \rangle$ for all $v_1, v_2 \in V$.
- (b) $|F(v)| = |v|$ for all $v \in V$.
- (c) If $\{v_1, \dots, v_n\}$ is an orthonormal basis in V , then $\{F(v_1), \dots, F(v_n)\}$ is an orthonormal basis in W .
- (d) There exists an orthonormal basis $\{v_1, \dots, v_n\}$ in V such that $\{F(v_1), \dots, F(v_n)\}$ is an orthonormal basis in W .

Solution. (b) $\implies$ (a). The magnitude condition can be written as

$$\langle F(v), F(v) \rangle = |F(v)|^2 = |v|^2 = \langle v, v \rangle.$$

Then, by using the linearity of F , we get

$$2\langle F(v), F(w) \rangle = |F(v+w)|^2 - |F(v)|^2 - |F(w)|^2 = |v+w|^2 - |v|^2 - |w|^2 = 2\langle v, w \rangle$$

for any $v, w \in V$.

(a) $\implies$ (c). The preservation of the dot product means that, given orthonormal basis $\{v_1, \dots, v_n\}$, we have

$$\langle F(v_i), F(v_j) \rangle = \langle v_i, v_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

which is exactly what it means for $\{F(v_1), \dots, F(v_n)\}$ to be an orthonormal basis for W .

(c) $\implies$ (d). All one has to do here is prove the existence of an orthonormal basis for V , since the result of (c) already tells us that the image of the basis will be an orthonormal basis for W . It is well-known that any finite-dimensional vector space with an inner product has an orthonormal basis, e.g. by using the Gram-Schmidt algorithm.

(d) $\implies$ (b). Let $\{v_1, \dots, v_n\}$ be an orthonormal basis of V such that $\{F(v_1), \dots, F(v_n)\}$ is an orthonormal basis of W . Let $v = a_1v_1 + \dots + a_nv_n$ be an arbitrary vector in V written with its coordinates in the basis $\{v_1, \dots, v_n\}$. Then

$$|v|^2 = \langle v, v \rangle = \langle a_1v_1 + \dots + a_nv_n, a_1v_1 + \dots + a_nv_n \rangle = \sum_{i=1}^n \sum_{j=1}^n \langle a_i v_i, a_j v_j \rangle = \sum_{i=1}^n a_i^2$$

and similarly,

$$|F(v)|^2 = \langle F(v), F(v) \rangle = \sum_{i=1}^n \sum_{j=1}^n \langle a_i F(v_i), a_j F(v_j) \rangle = \sum_{i=1}^n a_i^2,$$

where in both cases we reduce the sum to $\sum_{i=1}^n a_i^2$ by using the fact that $\langle v_i, v_j \rangle = \langle F(v_i), F(v_j) \rangle = 0$ if $i \neq j$ and 1 if $i = j$. Thus (d) implies that for any $v \in V$, $|v| = |F(v)|$.

Therefore we have shown that (b) implies (a) implies (c) implies (d) implies (b), so the four statements are equivalent. $\square$

Problem. 4.2.14. We say that a differentiable map $\varphi: S_1 \rightarrow S_2$ preserves angles when for every $p \in S_1$ and $v_1, v_2 \in T_p(S_1)$, we have

$$\cos(v_1, v_2) = \cos(d\varphi_p(v_1), d\varphi_p(v_2)).$$

Prove that φ is locally conformal iff it preserves angles.

Solution. Suppose first that φ is locally conformal. Then given $p \in S_1$, there exists a neighborhood $V \subseteq S_1$ of p such that $\varphi: V \rightarrow \varphi(V)$ is conformal. Then, as shown in the book, conformal maps preserve angles, so the equality in the problem statement is true.

Conversely, suppose that φ preserves angles. We wish to prove that for each $p \in S_1$, there is a neighborhood $V \subseteq S_1$ of p such that the restriction $\varphi: V \rightarrow \varphi(V)$ is conformal. Note that angles being preserved means

$$\frac{\langle v_1, v_2 \rangle}{|v_1||v_2|} = \frac{\langle d\varphi_p(v_1), d\varphi_p(v_2) \rangle}{|d\varphi_p(v_1)||d\varphi_p(v_2)|},$$

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Problem. 4.2.17. Consider a triangle on the unit sphere whose sides are segments of loxodromes and do not contain poles. Prove that the sum of the interior angles of such a triangle is π . $\square$

Solution. todo $\square$

Problem. 4.3.1. Show that if $\mathbf{x}$ is an orthogonal parametrization, that is, $F = 0$, then

$$K = -\frac{1}{2\sqrt{EG}} \left(\left(\frac{E_v}{\sqrt{EG}} \right)_v + \left(\frac{G_u}{\sqrt{EG}} \right)_u \right).$$

Solution. todo $\square$

Problem. 4.3.2.

Solution. todo $\square$

Problem. 4.3.3.

Solution. todo $\square$

Problem. 4.3.4.

Solution. todo $\square$

Problem. 4.3.8.

Solution. todo $\square$